AIscreenR as a Second Screener in Evidence Synthesis: A Proof of Concept Using a Health Technology Assessment of Herpes Zoster Vaccination

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Key summary

This proof of concept represents the first validation of an AI-assisted workflow for literature selection within a living evidence map for the Adult Vaccination Program in Norway, referred to as LEVAK (in Norwegian, *Levende Evidenskart for Vaksiner*). LEVAK is intended to be an interactive living evidence hub covering the three vaccines included in the national program: COVID-19, influenza, and pneumococcal vaccines. It will also monitor research on attitudes toward vaccination and interventions for increasing vaccination coverage and influencing public attitudes toward vaccination.

This report evaluates the R package `AIscreenR` for title and abstract screening, using a health technology assessment (HTA) of herpes zoster vaccination (HZV) as a benchmark [1]. Researchers at the Department of Infectious Disease Control and Vaccines at the Norwegian Institute of Public Health conducted the original manual screening of titles and abstracts. Their screening decisions serve as the reference standard for this proof of concept evaluation.

The unit of analysis for this validation consisted of 11,445 references retrieved by the HZV team from the databases included in the HTA [@HZV_FHI2026]. We prepared the reference library and developed prompts based on the selection criteria used in the HTA. To evaluate the performance of `AIscreenR`, we present a confusion matrix and report performance metrics, including accuracy and workload savings.

The screening results presented in this report are the product of an iterative process. An initial screening, referred to as HZV Screening 1, was conducted using a first draft of the prompt with GPT-4o-mini. Subsequently, a refined prompt was deployed in both GPT-4o-mini and GPT-5-mini to compare their performance. Based on the results of this iteration, we then conducted HZV Screening 2 using both models.

The best screening performance was obtained using GPT-5-mini with the fine-tuned prompt, as follows:

Key findings

AIscreenR correctly identified 87% of the references included by the HZV team. This performance exceeds the 75% threshold recommended in the literature for implementation within evidence synthesis workflows [2].

AIscreenR reduced the screening workload by nearly 82%. It safely identified and removed 9,337 references (82% of the total) that had also been excluded by the HZV team.

The LLMs tend to be over-inclusive, which allows researchers to safely double-check the data. AIscreenR included 1,972 references (17%) that were ultimately excluded by the HZV team. Consequently, the HZV team would have only needed to screen 2,079 out of the 11,445 total references to identify the 123 included studies.

Conclusion

AIscreenR should currently be utilized to prioritize screening rather than to replace human review entirely. This tool can safely be used as a second screener for title and abstract screening in an evidence synthesis, such as the one conducted by the HZV team. Fine tuning the prompts is crucial. The recall and specificity values align with the recommendations of Vembye et al. [2] for a trustworthy an valid use of AIscreenR in evidence synthesis. Further validation studies will help us better understand the potential of AIscreenR in prioritizing vaccine research.

Background

AIscreenR

AIscreenR [2] is an R package designed to streamline the title and abstract screening phase of evidence synthesis by deploying Large Language Models (LLMs) through the OpenAI API. AIscreenR provides a communication interface between researchers and large language models (LLMs). Researchers instruct the LLM to assess the eligibility of a study by using a prompt that mirrors the selection criteria. It is most commonly used to reduce the immense manual workload of evidence synthesis, particularly by serving as an efficient "second reviewer" that can accurately identify and remove irrelevant studies [2].

AIscreenR's performance was validated in 2025, using as a benchmark the screening decisions from 22 high-quality systematic reviews with different levels of complexity [2]. AIscreenR showed that the LLM performed on par with, and in some cases even better than, human reviewers in title and abstract screening [2]. The tool demonstrated high accuracy and reliability, especially when a "human-in-the-loop" strategy was employed.

Objective

To evaluate the performance of AIscreenR in screening references at the title and abstract level, using a health technology assessment on herpes zoster vaccination as the gold-standard comparative benchmark.

Methods

This proof-of-concept evaluation is conducted to better understand the performance of `AIscreenR` in title and abstract selection within evidence synthesis for vaccine research. We aim to deploy `AIscreenR` as the screener for relevant vaccine research to be mapped into a living evidence map (acronym: LEVAK). The methodological framework adheres to recent guidance presented by Cochrane [3].

As detailed in the full HZV report [1], an experienced librarian searched in electronic databases by an experienced librarian. Pairs of researchers manually and independently screened the studies and resolved disagreements on inclusion via consensus. The final list of included and excluded studies was provided by the HZV project leader (Eli Heen) in a RIS file. The following sections of the document outline the data cleaning and preparation, as well as the `AIscreenR` pipeline used to execute the screening.

We constructed a confusion matrix to display the performance of `AIscreenR` compared to the final screening decisions from the HZV report [1]. In evidence synthesis, the confusion matrix help us in [4]:

- Error Analysis: Identifying misclassification patterns.
- Bias Detection: Understanding the balance between False Negatives (FN) and False Positives (FP).
- LLM Selection: Comparing different models based on their screening performance.
- Fine-tuning: Evaluating how the LLMs respond to different prompting approaches.

The confusion matrix visualizes the following metrics:

• True Positives (TP): The number of references correctly included by `AIscreenR` (i.e., references also included by the HZV team).

• True Negatives (TN): The number of references correctly excluded by `AIscreenR` (i.e., references also excluded by the HZV team).

• False Positives (FP): References incorrectly included by `AIscreenR` (Type I errors), meaning references included by `AIscreenR` but excluded by the HZV team.

• False Negatives (FN): References incorrectly excluded by `AIscreenR` (Type II errors).

True negatives (TN) are important for reducing workload, as they represent the number of references that a review team can avoid screening by using AIscreenR. In contrast, false negatives (FN) represent the relevant references that would have been missed without human verification.

Based on the confusion matrix, we then calculated the following metrics:

• **Recall** (Sensitivity, also known as true positive rate): The proportion of truly relevant studies correctly included by `AIscreenR`. Recall is the most critical metric; a target of >90% ensures that study selection in LEVAK is safe and does not miss critical new evidence.

• **Specificity** (true negative rate): The proportion of truly irrelevant studies correctly excluded by `AIscreenR`. High specificity help researchers stay focused on relevant studies. If specificity is low, researchers will have to manually screen too many irrelevant papers (False Positives).

Data requirements

For the `AIscreenR` screening functions to operate successfully, the dataset must contain at least three specific variables: a unique identifier for each record (e.g., `studyid`), the article `title`, and the `abstract`.

It is also highly recommended to include a `human_code` variable containing previous screening decisions (0 for excluded, 1 for included) in order to validate the AI's performance.

Data quality

Before the screening phase, users must ensure data quality by filtering out records with missing abstracts and resolving any character encoding issues that could interfere with the API's token processing in `AIscreenR`.

In many academic workflows, the search results will most likely be saved in an EndNote library before being brought into R. To facilitate the import, users should export their EndNote library as an XML or RIS file, as these formats are the most stable for preserving the relationship between titles and abstracts.

Once exported, the `synthesisr::read_refs()` function can parse these files directly, maintaining the metadata structure required for the screening process. This function belongs to the `synthesisr` package, which is a specialized tool for importing, exporting, and manipulating bibliographic data in R.

HZV use case

On 30 January 2026, the research librarian (Marita Heintz) provided the literature search results in a RIS file (.ris) format. The results of the title and abstract screening for the original Herpes Zoster vaccine HTA were not available. Instead, Eli Heeen and Lene Juvet provided Jose Meneses with the final list of included studies and those excluded at the full-text stage.

To reconstruct the dataset for validation, Jose Meneses manually coded these references in EndNote to establish the following classifications:

- **Title and Abstract Screening (TA)**
  - Included references (`Included_TA`)
  - Excluded references (`Excluded_TA`)
- **Full-Text Screening (FT)**
  - Included references (`Included_FT`)
  - Excluded references (`Excluded_FT`)

After manually coding the `Included_TA` references in EndNote, we reconstructed the `Excluded_TA` set by subtracting the `Included_TA` references from the complete library of references. To replicate this in EndNote, follow these steps:

1. Navigate to **Groups** > **Create From Groups**.
2. Name the new group (e.g., `Excluded_TA`).
3. Under "Include References in," select the **All References** group.
4. Set the operator to **NOT** and select the `Included_TA` group in the dropdown.
5. Click **Create**.

Human_code variable

The `human_code` variable allows us to validate `AIscreenR` 's performance, as it contains the screening decisions made by the HZV team (i.e., the human reviewers).

We manually created a new field in EndNote to assign human assessments to each reference. This `human_code` variable must be 0 for `Excluded_TA` and 1 for `Included_TA`. To add this variable to each reference in our RIS file, we must first create the custom field in EndNote using the following steps:

Step 1: Create the Field in Your References

To make a tag "global," you must first ensure the field exists within your database records.

1. Go to **Edit** > **Preferences** (Windows) > Select **Reference Types** from the menu > **Modify Reference Types**.
2. Find one of the empty columns (such as `Custom 1`) and type the name you want (e.g., *human_code*).
3. Click **OK**. All your references will now have this new field available.

Step 2: Bulk Fill the Field

If you want this field (tag) to appear across all existing references without typing them one by one:

1. Select all your references (**Ctrl+A**).
2. Go to the **Library** menu > **Change/Move/Copy Fields**.
3. In the **Change Fields** tab, select the field you just named (e.g., *Custom 1*).
4. Choose the option **Replace whole field with:** and type the text you want to appear (e.g, 1 for `Included_TA`).
5. Press **OK**.
6. Repeat these steps for the `Excluded_TA`.

Step 3: Configure the RIS Export Format

EndNote does not export custom fields by default. You must instruct it to associate your new field with a RIS tag (such as `U1`).

1. Go to **Edit > Output Styles > Edit "RefMan (RIS) Export"**.
2. In the menu on the left, click on **Templates**.
3. Locate the reference type you use most (e.g., *Journal Article*).
4. You will see a list of tags like `AU  - Author`. Go to the end of the list and manually add:
   - Type `U1  -` and then click the **Insert Field** button to select your custom field (e.g., *Custom 1*).
5. **Save your changes:** Go to **File > Save As** and give it a name like `Custom RIS Export`.

Step 4: Exporting the edited RIS file
Ensure that when you clicked **File > Export** in EndNote, you actually select the new "`Custom RIS Export`" style, rather than the default "RefMan (RIS) Export, to ensure the `U1` field is included in the final file.

Adjustment to references (optional)
We restricted the database to the variables required by `AIscreenR`: ID, title, abstract, and `human_code`. The `human_code` variable is essential only when human screening results are available, as it enables a direct comparison between `AIScreenR`'s performance and the decisions made by human reviewers.

AIscreenR workflow

The following is a step-by-step description of the workflow:

Package and Data upload
It is practical to centralize all packages and functions used for the evaluation and reporting into a dedicated code chunk.

```
#install.packages("rsconnect")
library(rsconnect)

# Load packages and libraries
library(AIscreenR)
library(writexl)
library(dplyr)
library(synthesisr)
library(tibble)
library(revtools)
library(flextable)
library(tinytex)
library(tidyr)
library(ggplot2)
library(gt)
library(webshot2)
library(packrat)
```

```
library(rsconnect)
library(chromote)
library(readxl)
library(future)

#devtools::install_github("MikkelVembye/AIscreenR")  #  most  recent  pakcage
updates

#Load data of the project
load(".RData")
```

Set API Key

AIscreenR uses an API key to connect to a Large Language Model (LLM), in this case, the OpenAI API to connect to ChatGPT. After obtaining your API key (which is private and confidential), you can save it using the following code:

```
usethis::edit_r_environ() # Run this code to open your .Renviron file and
write CHATGPT_KEY=your_key. Thereafter, close and save the .Renviron file and
restart RStudio (ctrl + shift + F10). From now on, the AIscreenR functions will
automatically use get_api_key() to retrieve your API key from your R environment.

get_api_key() # To check the API Key was saved successfully.
```

Importing the dataset

To import references for AIscreenR, you typically utilize the synthesisr or revtools packages to convert standard bibliographic files—such as .ris, .bib, or .txt—into an R-compatible data frame. The process begins by reading the raw file into R, which generates a table where each row represents a unique study [2].

We import the references that will be screened by AIscreenR.

```
confusion_helvetesild3_values<-                                read_excel("C:/
Users/JOFM/OneDrive  -  Folkehelseinstituttet/Research  projects/FHI/Smittevern/
VVP/Helvetesild_testing/confusion_helvetesild3_excel.xlsx") #main file with the
references (search results)

HZV_references<-confusion_helvetesild3_values %>%
  select(ID, human_code, title, abstract, studyid, promptid, prompt, n)

View(HZV_references) #inspecting the references
save(HZV_references, file = "HZV_references.RData")
save.image(file = ".RData")
```

Missing titles check

Missing titles: Use the following code to ensure no references are missing a title.

```
sum(is.na(HZV_references$title)) # check missing titles
```

```
[1] 0
```

All the references have a title (No titles missing).

Missing abstract check

The abstract is the primary source of information used by the LLM to determine whether a study meets the inclusion criteria. `AIscreenR`, independent of the specific GPT model employed, tends to exclude references by default when information is insufficient. If a study without an abstract is actually relevant, it may result in a False Negative (key evidence is omitted from the synthesis).

Furthermore, the validation metrics for `AIscreenR` assume the LLM has access to the same data as a human reviewer. Empty abstract fields artificially lower the performance scores, leading to biased conclusions regarding the tool's validation.

The abstracts in the database show "No information" instead of "NA". Use this code to standardize the missing values to "NA":

```
# # Verify the "No information" items.
# sum(HZV_references$abstract == "No information", na.rm = TRUE)

# # Convert "No information" to a real "NA".
# HZV_references$abstract[HZV_references$abstract == "No information"] <- NA
```

We can check the number of missing abstracts now:

```
sum(is.na(HZV_references$abstract)) # check missing abstracts
```

```
[1] 0
```

Abstracts are missing for 1,616 references, representing 14% of the 11,445 total entries in the EndNote library.

HZV screening 1

This is our first, exploratory screening.

Prompt (HZV screening 1)

The prompt is the set of instructions that the LLM follows to determine the relevance of each study. It must mirror the selection criteria, defining the key inclusion and exclusion criteria as clearly as possible. To ensure high accuracy, the prompt should be detailed and thorough.

You can draft the prompt in either R or Word before integrating it into the pipeline.

```
# HZV_prompt <- "Role: You are an expert systematic review screener specializing
in vaccinology and epidemiology.
# Task: Evaluate whether a study meets the following specific inclusion criteria.
# Population: Adults aged fifty years or older. Include participants regardless
of previous vaccination history, history of herpes zoster, or presence of
comorbidities (e.g., cardiovascular disease, Alzheimer's).
# Intervention: Recombinant Zoster Vaccine (RZV) (e.g., Shingrix, HZ/su) or
Zoster Vaccine Live (ZVL) (e.g., Zostavax).
# Comparison/Control: Placebo, no vaccine, or an alternative vaccine (including
head-to-head comparisons between RZV and ZVL).
# Study Design: Included: RCTs, non-randomized studies, observational studies
(prospective/retrospective cohort, registry-based, quasi-experimental, case-
control, nested case-control), and post-marketing safety evaluations.
# Secondary Research: Include systematic reviews, meta-analyses, and literature
reviews.
# Excluded: Case studies and case series.
# Evaluation Rule: Does the study meet all the inclusion criteria? Only include
the study if every criterion is satisfied."
```

Screening command

```
#HZV_screening <- tabscreen_gpt(
  #data = HZV_references,
  #prompt = HZV_prompt,
  #studyid = ID,
  #title = title,
  #abstract = abstract,
  #model = "gpt-5-mini",
  #reps = 1,
  #decision_description = TRUE
#)
```

Results (HZV screening 1)

Exploring the screening results

Print the results of the screening done by `AIscreenR`. Check the following variables:

- **decision_gpt:** the raw gpt decision - either "1", "0", "1.1" for inclusion, exclusion, or uncertainty.
- **detailed_description:** the detailed description of the given decision made by OpenAI's GPT API models.
- **decision_binary:** the binary gpt decision, that is 1 for inclusion and 0 for exclusion.

```
HZV_screening <- readRDS("HZV_screening.rds") # load the screening results

# Preview of the screening results data.
view(HZV_screening$answer_data) # inspect the data

# Distribution of LLM-generated screening decisions
HZV_screening$answer_data %>%
  count(decision_binary)

#Saving the results file
saveRDS(HZV_screening, "HZV_screening.rds")
```

`AIscreenR` excluded 10,227 references and included 1,214.

We found that *four* references were marked as 'NA' by the software. We checked these manually and reclassified them as excluded (0) as they are not deemed relevant to the study.

```
#Assign 0 to the references with NA (n=4)
HZV_screening$answer_data$decision_binary[is.na(HZV_screening$answer_data$decision_binary)]<-0

#Check
HZV_screening$answer_data %>%
  count(decision_binary)
```

After cleaning the data, `AIscreenR` excluded 10,231 references and included 1,214 references Table 1.

```
# Data wrangling_row data (first table)
HZV_screening_rawtable <- HZV_screening$answer_data %>%
  count(decision_binary) %>%
  mutate(
    Decision = if_else(decision_binary == 1, "Included", "Excluded"),
    Percentage = n / sum(n)
  ) %>%
  select(Decision, n, Percentage)
view(HZV_screening_rawtable)

# Saving the object in R and also as Excel
saveRDS(HZV_screening_rawtable, "HZV_screening_rawtable.rds")
writexl::write_xlsx(HZV_screening_rawtable, "HZV_screening_rawtable.xlsx")
```

Table 1: AIScreenR screening results

AIScreenR screening results

Distribution of AIScreenR Screening Results

Decision	Frequency (n)	Percentage (%)
Excluded	10231	89.4%
Included	1214	10.6%

Note: Four references returned an error in AIScreenR; these were manually reviewed and categorized as excluded

Now that the data are clean with unique values of either included and excluded by `AIscreenR`, we can proceed to the analysis.

Analyzing the results

`AIscreenR` (and other AI tools) is seen as a diagnostic tool, detecting relevant references for the review question and excluding irrelevant ones. There are two primary entries: the binary inclusion/exclusion decisions made by the HZV team (human reviewers) and those made by `AIscreenR`. This comparison results in a 2x2 table, which we refer to as a 'Confusion Matrix' [4].

Confusion matrix

To prepare the key metrics for the confusion matrix, we retrieve the raw data (the distribution of decisions made by both `AIscreenR` and humans), organize these into a data frame, and then generate the confusion matrix table.

1. Retrieve and prepare the raw data.

```
# Preparing the raw data for the metrics:
HZV_confusion <- table(
  Human = HZV_screening$answer_data$human_code,
  AI = HZV_screening$answer_data$decision_binary
)
```

2. Organize the data in a data frame.

```
# Converting the raw data of the confusion matrix to a data frame (mandatory
step in dplyr)
HZV_confusion_df <- as.data.frame(HZV_confusion)
colnames(HZV_confusion_df) <- c("human_code", "decision_binary", "n")

HZV_confusion_final <- HZV_confusion_df %>% # using the dataframe (df)
  mutate(
    Human = if_else(human_code == 1, "Included", "Excluded"),
```

```
    AIscreenR = if_else(decision_binary == 1, "Included", "Excluded")
  ) %>%
    pivot_wider(id_cols = Human, names_from = AIscreenR, values_from = n)
```

3. Plotting the confusion matrix table.

The evaluation of the HZV screening process (n = 11,445 references from the searches) is shown in Table 2.

Table 2: Confusion matrix AIscreenR vs HZV team

Confusion matrix AIscreenR vs HZV team

Performance analysis in study selection

	AIscreenR	
Human reviewers	Included	Excluded
Included	75	48
Excluded	1138	10171
Total	1,213	10,219

Interpreting the results:

- **True Negatives (TN)**= 10,171 references (88.9%) were correctly excluded by both the HZV team and `AIscreenR`. In other words, `AIscreenR` safely removed 89% of the irrelevant records.
- **False Negatives (FN)** = 48 references (0.4%) included by the HZV team but missed by `AIscreenR`.
- **True Positives (TP)** = 75 references. These are the relevant studies correctly identified by both HZV team and `AIscreenR`.
- **False Positives (FP)** = 1,138 references were included by `AIscreenR` but excluded by the HZV team. This represents the *"residual manual workload"* (i.e., the 1,138 papers the review team still needs to check manually to ensure they are truly irrelevant).

Key performance metrics

Based on the key performance metrics in the confusion matrix, we can now estimate the accuracy and the recall.

Recall (Sensitivity)

Recall represents the proportion of truly relevant references that were correctly identified and included by `AIscreenR`. This is critical metric as it measures the `AIscreenR`'s ability to find "all the needles in the haystack." A high recall ensures that no relevant studies are missed during the screening process.

Accuracy

Accuracy is the proportion of all decisions made by `AIscreenR` (both inclusions and exclusions) that were correct. While informative, accuracy can sometimes be misleading in systematic reviews if the number of irrelevant papers vastly outweighs the relevant ones (class imbalance).

The `screen_analyzer` function in `AIscreenR` calculates both recall and accuracy Table 3.

Table 3: Performance metrics of AIscreenR results

Performance metrics of AIscreenR results Comparative performance of AIscreenR in screening references at the title and abstract level on herpes zoster vaccination as the gold-standard comparative benchmark		
gpt-4o-mini	Recall (Sensitivity)	Specificity
gpt-4o-mini	61.0%	89.9%

Specificity

Specificity measures the proportion of truly irrelevant references that were correctly excluded by `AIscreenR`. Unlike Recall that focuses on finding what is relevant, Specificity focuses on accurately "clearing out" the noise. High specificity reduces the number of irrelevant references that researchers must manually double-check.

Workload Savings

Workload savings quantifies the efficiency gain provided by `AIscreenR`, expressed as the percentage of the total references that the review team no longer needs to screen manually because `AIscreenR` correctly excluded. This metric directly reflects the time and resource reduction achieved by implementing `AIscreenR`.

?@tbl-keymetrics presents the key performance metrics of `AIscreenR` in screening the studies compared to the HZV team.

Performance metric	Value	Interpretation
Recall (Sensitivity)	61.0%	The most vital metric for quality. It shows that `AIscreenR` captured 61% of the relevant references. The remaining 39% (the 48 False Negatives) were missed.
Accuracy	89.5%	The overall proportion of correct decisions (both inclusions and exclusions) made by `AIscreenR`. While high, accuracy can be misleading in imbalanced datasets.

Performance metric	Value	Interpretation
Specificity	89.9%	The ability of AIscreenR to correctly identify and discard irrelevant studies. A 90% specificity is excellent for significantly narrowing down a large library.
Workload savings	89.4%	The "Efficiency Gain." It measures the percentage of the total library that the HZV team was able to skip screening entirely thanks to AIscreenR.

Conclusions (HZV screening 1)

- AIscreenR should be used to prioritize screening rather than replacing human review entirely.
- AIscreenR identified 61% of the references included by the HZV team, meaning it overlooked 48 relevant studies. This recall should be higher (at least 75% before full implementation in evidence synthesis workflows).
- AIscreenR resulted in a workload reduction of nearly 90% during the tittle and abstract screening of the 11,445 references retrieved from the databases.
- The HZV team could have safely excluded 90% of the total references by using AIscreenR.

False Negative Analysis

To address the low recall observed in the initial AIscreenR screening, we must analyze the False Negatives (FNs) to identify potential improvements for the prompt. The FNs are those relevant studies that AIscreenR excluded but the humans had included.

A refined prompt should, in theory, improve AIscreenR's recall by providing the LLM with clearer inclusion and exclusion boundaries.

Select the False Negatives

The unit of analysis at this stage consists of all False Negatives identified in the screening confusion matrix. We must create a dedicated variable containing these FNs to systematically inspect the exclusion reasons provided by AIscreenR Table 2.

```
# Create the groups in the confusion matrix: TP, TN, FN, FP
HZV_screening_excel                    <-                    read_excel("C:/
Users/JOFM/OneDrive - Folkehelseinstituttet/Research projects/FHI/Smittevern/
VVP/Helvetesild_testing/HZV_screening_excel.xlsx")

HZV_confusion<- HZV_screening_excel%>%  # new object
  mutate(result_type = case_when(
    human_code == 1 & decision_binary == 1 ~ "True Positive (TP)",
```

```
    human_code == 0 & decision_binary == 0 ~ "True Negative (TN)",
    human_code == 0 & decision_binary == 1 ~ "False Positive (FP)",
    human_code == 1 & decision_binary == 0 ~ "False Negative (FN)"
  ))
view(HZV_confusion)
saveRDS(HZV_confusion, "HZV_confusion.rds")

# Extract the FNs from "HZV_confusion"
HZV_FN<- HZV_confusion%>%
  filter(result_type== "False Negative (FN)")
view(HZV_FN)
# This is the new variable containing all the False Negatives(FNs).

writexl::write_xlsx(HZV_FN, "HZV_FN.xlsx")
saveRDS(HZV_FN, "HZV_FN.rds")
```

With the False Negatives now stored in an object called 'HZV_FN', we will export this dataset to review the exclusion reasons provided by `AIscreenR` (column "detailed_description"). This analysis will directly inform the necessary refinements to our screening prompt.

Tuning the prompt

Analysis of the exclusion reasons revealed that `AIscreenR` incorrectly excluded references that did not specify the vaccine type, did not limit participants to a specific age group, focused on vaccine safety and studies on varicella/chickenpox vaccines.

The revised and tuned prompt is now saved as a new variable in our R environment.

```
HZV_prompt_tuned<-"Role: You are an expert systematic review screener
specializing in vaccinology and epidemiology.
Task: Evaluate whether a study meets the following specific inclusion criteria.
Population: Adults of any age (18 years and older), with a specific focus on
including middle-aged, older adults, and the elderly (e.g., 60–70 years, 80+
years, and beyond). Include individuals regardless of:
Vaccination History: Including those with or without a previous history of
vaccination (e.g., varicella/chickenpox).
Medical History: Including those with or without prior episodes of herpes zoster.
Health Status: Including participants with or without existing comorbidities
or chronic conditions. This encompasses healthy populations as well as
participants with conditions such as cardiovascular disease, organ or stem cell
transplantation, Alzheimer's disease, or other immunocompromising factors.
Intervention: Studies evaluating the effectiveness and safety of either the
Recombinant Zoster Vaccine (RZV; e.g., Shingrix, HZ/su), the Zoster Vaccine Live
(ZVL; e.g., Zostavax), or unspecified zoster vaccines, such as those evaluated
in national vaccination programs.
```

```
Comparison/Control: Placebo, no vaccine, or alternative vaccines (including head-to-head comparisons between RZV and ZVL).
Study Design: Randomized controlled trials (RCTs), non-randomized studies, and observational studies (prospective/retrospective cohort, population-based, registry-based, quasi-experimental, case-control, and nested case-control). Systematic reviews, meta-analyses, and post-marketing safety evaluations are also eligible.
Exclude the following: Commentaries, narrative reviews, case studies, and case series.
Evaluation Rule: Does the study meet all the inclusion criteria? Only include the study if every criterion is satisfied.
If you are uncertain about the inclusion of a study or if the study shows signs that the full text might meet the inclusion criteria,
choose INCLUDE to minimize the risk of missing relevant studies."
```

Running the screening

```
# Screening command
#HZV_FN_screening <- tabscreen_gpt(
  #data = HZV_FN,
  #prompt = HZV_prompt_tuned,
  #title = title,
  #abstract = abstract,
  #studyid = ID,
  #model = "gpt-4o-mini",
  #decision_description = TRUE,
  #max_tries = 10,
  #reps = 1
  #)
```

Analyzing the results

To evaluate the impact of the tuned prompt, we measured the reduction in False Negatives (FNs). This is calculated by subtracting the new FNs (tuned) from the FNs in the initial screening. This equals the prompt's effectiveness in capturing previously missed relevant references, thereby improving recall.

```
#HZV_FN_screening$answer_data%>%  #view the results
  #count(decision_binary)
```

- The previous number of FNs in HZV_confusion_final was 48.
- HZV_tuned_screening = `AIscreenR` using GPT-4o-mini included 29 (TPs) references and excluded 19 references (FNs).

- FN reduction = 29 references (60% FN reduction rate).

The fine-tuned prompt reduced the FNs rate by 60%.

False Negative analysis using GPT-5-mini

We evaluate the comparative performance of OpenAI's recent model, GPT-5-mini, which is optimized for high-volume requests and offers greater speed than GPT-4o-mini. According to OpenAI's technical documentation [5], GPT-5-mini excels in efficiency, featuring a 400K token context window, which is a significant increase over the 128K window found in GPT-4o-mini. Newer models will be tested in future updates of this work.

To validate the performance of GPT-5-mini in controlling false negatives (FNs), we re-ran the original prompt on the identified FN cases and switched the model variable to GPT-5-mini, and compared the results.

```
#HZV_FN_gpt5_screening <- tabscreen_gpt(
 #data = HZV_FN,
  #prompt = HZV_prompt_tuned,
  #studyid = ID,
  #title = title,
  #abstract = abstract,
  #model = "gpt-5-mini",  ## changing the model **
  #reps = 1,
  #decision_description = TRUE
  #)

# Distribution of LLM-generated screening decisions
HZV_FN_gpt5_screening <- readRDS("HZV_FN_gpt5_screening.rds")

HZV_FN_gpt5_screening$answer_data %>%
  count(decision_binary) #Checking the raw numbers

view(HZV_FN_gpt5_screening$answer_data) # inspect the data

#Saving the results file
saveRDS(HZV_FN_gpt5_screening, "HZV_FN_gpt5_screening.rds")
```

GPT-5-mini included 33 FN references (69% FN reduction rate), relative to the 29 references included by GPT-4o-mini (60% FN reduction rate). This represents a 9% screening performance gain.

HZV screening 2

The fine-tuned prompt, which improved screening performance during False Negative Analysis, can now be used to re-run the screening process. This iteration is expected to enhance the overall recall of `AIscreenR` by reducing the False Negatives.

The impact of the fine-tuned prompt will be evaluated using both gpt-4o-mini and gpt-5-mini.

HZV screening 2 using gpt-4o-mini

Running the screening using the fine-tuned prompt:

```
#HZV_screening2 <- tabscreen_gpt(
  #data = HZV_references,     #The same set of references.
  #prompt = HZV_prompt_tuned, #This is the tuned prompt.
  #studyid = ID,
  #title = title,
  #abstract = abstract,
  #model = "gpt-4o-mini",
  #reps = 1,
  #decision_description = TRUE
  #)
```

In **HZV screening 2 using gpt-4o-mini**, `AIscreenR` included 1,368 and excluded 10,077 references.

```
HZV_screening2 <- readRDS("HZV_screening2.rds")

#Saving the results file
saveRDS(HZV_screening2, "HZV_screening2.rds")

# Preview of the screening results data.
view(HZV_screening2$answer_data) # inspect the data

# Distribution of LLM-generated screening decisions
HZV_screening2$answer_data %>%
  count(decision_binary)

#Assign 0 to the references with NA (n=4)
HZV_screening2$answer_data$decision_binary[is.na(HZV_screening2$answer_data$decision_binary)]<-0

#Check
HZV_screening2$answer_data %>%
  count(decision_binary)

writexl::write_xlsx(HZV_screening2$answer_data, "HZV_screening2.xlsx") # Export
to Excel
```

Confusion matrix

1. Preparing the raw data for the confusion matrix.

```
# Preparing the raw data for the metrics:
HZV_confusion2 <- table(
  Human = HZV_screening2$answer_data$human_code,
  AI = HZV_screening2$answer_data$decision_binary
)
print(HZV_confusion2)
view(HZV_confusion2)
colnames(HZV_confusion2)
```

2. Organizing the data in a data frame.

```
# Converting the raw data of the confusion matrix to a data frame (mandatory
step in dplyr)
HZV_confusion2_df <- as.data.frame(HZV_confusion2)
colnames(HZV_confusion2_df) <- c("human_code", "decision_binary", "n")

HZV_confusion2_final <- HZV_confusion2_df %>% # using the dataframe (df)
  mutate(
    Human = if_else(human_code == 1, "Included", "Excluded"),
    AIScreenR = if_else(decision_binary == 1, "Included", "Excluded")
  ) %>%
    pivot_wider(id_cols = Human,names_from = AIScreenR,values_from = n)
View(HZV_confusion2_final)
```

3. Plotting the confusion matrix

The evaluation of the **HZV screening 2 using gpt-4o-mini** (n = 11,445 references from the searches) is shown in Table 4.

Table 4: Confusion matrix AIscreenR vs HZV team - tuned prompt using gpt-4o-mini

Confusion matrix AIscreenR vs HZV team - tuned prompt

Performance analysis in study selection

	AIscreenR	
Human reviewers	Included	Excluded
Included	96	27
Excluded	1271	10038
Total	1,367	10,065

Interpreting the results of **HZV screening 2 using gpt4-o-mini**:

- **True Negatives (TN)**= 10,038 references (88%) were correctly excluded by both the HZV team and `AIscreenR`. In other words, `AIscreenR` safely removed 89% of the irrelevant records.

- **False Negatives (FN)** = 27 references (0.2%) included by the HZV team but missed by `AIscreenR`.
- **True Positives (TP)** = 96 references. These are the relevant studies correctly identified by both HZV team and `AIscreenR`.
- **False Positives (FP)** = 1,271 references were included by `AIscreenR` but excluded by the HZV team. This represents the residual manual workload (i.e., the 1,271 papers the review team still needs to check manually to ensure they are truly irrelevant).

Key performance metrics

We estimate the accuracy and the recall of the fine-tuned screening **?@tbl-keymetricstuned**.

Performance metric	Value	Interpretation
Recall (Sensitivity)	78%	`AIscreenR` correctly identifies 78% of actually included references.
Accuracy	89%	Overall, `AIscreenR` correctly classifies 89% of all references.
Specificity	89%	`AIscreenR` correctly identifies 89% of actually excluded references.
Workload savings	89%	The HZV team would have skipped the manual screening of 10,038 references (89%) by using `AIscreenR`.

Full screening using gpt-5-mini

Running the screening using the fine-tuned prompt:

```
#HZV_screening_5gpt <- tabscreen_gpt(
  #data = HZV_references,
  #prompt = HZV_prompt_tuned,
  #studyid = ID,
  #title = title,
  #abstract = abstract,
  #model = "gpt-5-mini",
  #reps = 1,
  #decision_description = TRUE
#)
```

Inspecting the screening results:

```
HZV_screening_5gpt <- readRDS("HZV_screening_5gpt.rds")
# Preview of the screening results data.
view(HZV_screening_5gpt$answer_data) # inspect the data
```

```
# Distribution of LLM-generated screening decisions
HZV_screening_5gpt$answer_data%>%
  count(decision_binary)

#Saving the results file
saveRDS(HZV_screening_5gpt, "HZV_screening_5gpt.rds")
```

In **HZV screening 2 using gpt-5-mini**, `AIscreenR` excluded 9,366 references and included 2,079. Of note, *two* references were marked as 'NA' by the software. We checked these manually and reclassified them as excluded (0) as they are not deemed relevant to the study.

Confusion matrix

1. Preparing the raw data for the confusion matrix.

```
# Preparing the raw data for the metrics:
HZV_confusion2_gpt5 <- table( Human = HZV_screening_5gpt$answer_data$human_code,
AI = HZV_screening_5gpt$answer_data$decision_binary)
print(HZV_confusion2_gpt5)
colnames(HZV_confusion2_gpt5)
```

2. Organizing the data in a data frame.

```
# Converting the raw data of the confusion matrix to a data frame (mandatory
step in dplyr)
HZV_confusion2_gpt5 <- as.data.frame(HZV_confusion2_gpt5)
colnames(HZV_confusion2_gpt5) <- c("human_code", "decision_binary", "n")

HZV_confusion2_gpt5_final <- HZV_confusion2_gpt5 %>% # using the dataframe (df)
  mutate(
    Human = if_else(human_code == 1, "Included", "Excluded"),
    AIscreenR = if_else(decision_binary == 1, "Included", "Excluded")
  ) %>%
    pivot_wider(id_cols = Human,names_from = AIscreenR,values_from = n)
View(HZV_confusion2_gpt5_final)
```

3. Plotting the confusion matrix

The evaluation of the **HZV screening 2 using gpt-5-mini** is shown in Table 5.

Table 5: Confusion matrix AIscreenR vs HZV team - tuned prompt using gpt-5-mini

Confusion matrix AIscreenR vs HZV team - tuned prompt using gpt-5-mini

Performance analysis in study selection

	AIscreenR	
Human reviewers	Included	Excluded
Included	107	16
Excluded	1972	9337
Total	2,079	9,353

Interpreting the results of **HZV screening 2 using gpt-5-mini**:

- **True Negatives (TN)**= 9,337 references (82%) were correctly excluded by both the HZV team and `AIscreenR`. In other words, `AIscreenR` safely removed 82% of the irrelevant records.
- **False Negatives (FN)** = 16 references (0.14%) included by the HZV team but missed (or wrongly excluded) by `AIscreenR`.
- **True Positives (TP)** = 107 references. These are the relevant studies correctly identified by both HZV team and `AIscreenR`.
- **False Positives (FP)** = 1,972 references (17%) were included by `AIscreenR` but excluded by the HZV team. This represents the *"residual manual workload"* (i.e., the 1,972 papers the review team still needs to check manually to ensure they are truly irrelevant).

Key performance metrics (HZV screening 2 using gpt-5-mini)

We estimate the accuracy and the recall of the fine-tuned screening using `screen_analyzer` function in `AIscreenR`

Performance metric	Value	Interpretation
Recall (Sensitivity)	87%	`AIscreenR` correctly identifies 87% of actually included references.
Accuracy	83%	Overall, `AIscreenR` correctly classifies 83% of all references.
Specificity	83%	`AIscreenR` correctly identifies 83% of actually excluded references.
Workload savings	82%	The HZV team would have skipped the manual screening of 9,337 references (82%) by using `AIscreenR`.

Conclusion

The comparison of GPT-4o mini and GPT-5 mini using the fine-tuned prompt in both models revealed higher recall for GPT-5 mini (87%) compared with GPT-4o mini (78%), whereas specificity was slightly higher for GPT-4o mini (89%) than for GPT-5 mini (83%). This also translated into greater workload savings with GPT-4o mini (89% vs. 82% with GPT-5 mini).

This proof of concept suggests that LLMs can be safely used to prioritize relevant studies during title and abstract screening of HZV vaccine research. We will use GPT-5 mini in our literature selection pipelines as part of the Living Evidence Map of Vaccine Research (LEVAK). The same pipeline presented in this document will be used for subsequent validations in the diseases included in LEVAK: influenza, COVID-19, and pneumococcal disease.

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